

# 07 — Mock paper

All 13 of the professor's board questions, plus theory prompts on the lectured topics · marks unknown (no hand-out or past paper yet) · attempt closed-book, then mark with the answers

## HOW TO USE THIS

Do it in one sitting without opening 01–06. Write every step: the working earns marks, not just the final number. Mark each item clean / with struggle / wrong, and schedule it as in 00.

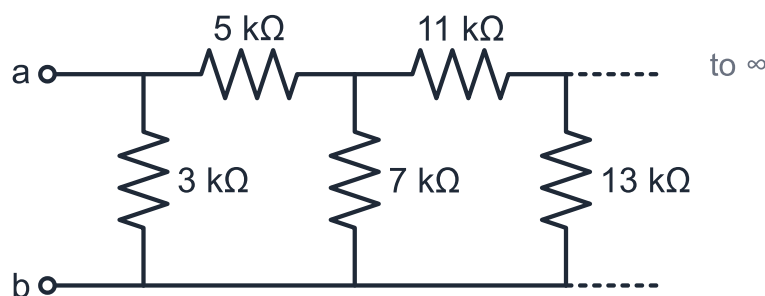
## Section A — Theory

Prompts on the topics the professor lectured. No past paper has been seen yet, so the *wording* of these is not the professor's.

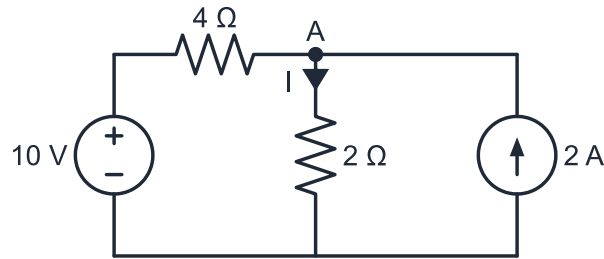
1. Classify circuit elements. Distinguish active and passive elements, with examples. Draw the symbols of the four dependent sources.
2. Derive  $v = L di/dt$  from  $\psi = N\phi$ . Explain why an inductor does not allow a sudden change of current.
3. State the superposition theorem. List its limitations.
4. Write the procedure for Thévenin's theorem. How is  $R_{Th}$  found when the network contains dependent sources?
5. State Norton's theorem. Give its procedure and draw the equivalent model.
6. Derive the condition for maximum power transfer to a resistive load. State the condition for a complex load.
7. Explain  $t = 0^-$ ,  $0^+$  and  $\infty$ . Write the general first-order response and the time constants of RL and RC circuits.

## Section B — Numericals (the professor's board questions)

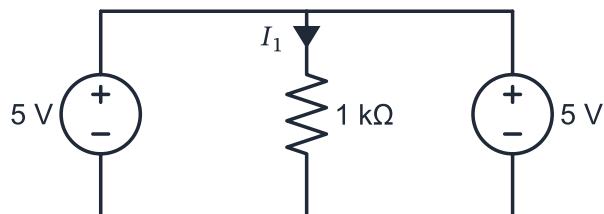
- B1** Class p.4 15 V is applied across 3 k $\Omega$  and 6 k $\Omega$  in series. Find  $V_1$  and  $V_2$ .
- B2** Class p.5 A 12 A current source feeds 3 k $\Omega$ , 6 k $\Omega$  and 4 k $\Omega$  in parallel. Find  $I_1$ ,  $I_2$ ,  $I_3$ .
- B3** Class p.5  $v = 10 \sin(2000t + 30^\circ)$  V across 3 k $\Omega$  and 7 k $\Omega$  in series. Find  $v_1$ ,  $v_2$ .
- B4** Class p.1 Find  $R_{eq}$  of the infinite ladder 3 k $\Omega$  shunt, 5 k $\Omega$  series, 7 k $\Omega$  shunt, 11 k $\Omega$  series, 13 k $\Omega$  shunt, ...



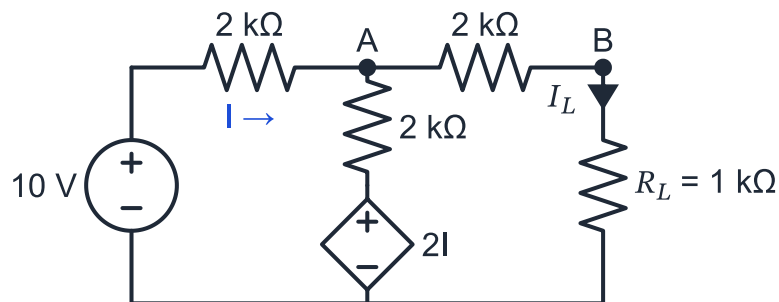
- B5** Class p.7 Calculate the current  $I$  using superposition.



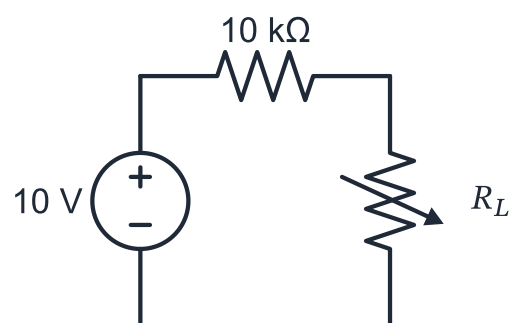
**B6** Class p.9 Calculate the current  $I_1$  using superposition. Comment on the result.



**B7** Class p.11 Determine the load current using Thévenin's theorem. Verify with Norton's theorem.



**B8–B9** Class p.16–17 For the circuit below: (a) the value of  $R_L$  for maximum power; (b) the total power delivered by the source; (c) the power consumed by  $R_{Th}$ ; (d) the efficiency; (e) the voltage across the load; (f) the relation between load power and total power, and between load voltage and source voltage.



**B10** Class p.17 10 V source with 5 kΩ: determine  $R$  for maximum power, the maximum power, and the efficiency.

**B11** Class p.19–20  $V$ , switch closing at  $t = 0$ ,  $R$ . Find  $I(0^-)$ ,  $I(0^+)$ ,  $I(\infty)$ .

**B12** Class p.21  $V$ , switch closing at  $t = 0$ ,  $R$  and  $L$  in series. Find  $I_R$ ,  $I_L$  at  $0^-$ ,  $0^+$ ,  $\infty$ . Then write  $i(t)$  for  $V = 10$  V,  $R = 1$  kΩ,  $L = 1$  mH.

**B13** Class p.22 10 V, a switch opening at  $t = 0$ ,  $1\text{ k}\Omega$  and  $1\text{ mH}$  in series. Find  $I_L(0^-)$  and  $I_L(0^+)$ . What happens to the inductor's current after the switch opens?

## Answers

**Section A:** model answers are the **Exam form** sections: 01 (A1, A2), 03 (A3), 04 (A4, A5), 05 (A6), 06 (A7).

|              | Answer                                                                                                                                                                                        | Where it's taught |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>B1</b>    | $V_1 = 5 \text{ V}$ , $V_2 = 10 \text{ V}$                                                                                                                                                    | 02 §1             |
| <b>B2</b>    | 5.33 A, 2.67 A, 4 A                                                                                                                                                                           | 02 §2             |
| <b>B3</b>    | $v_1 = 3 \sin(2000t + 30^\circ) \text{ V}$ , $v_2 = 7 \sin(2000t + 30^\circ) \text{ V}$                                                                                                       | 02 §3             |
| <b>B4</b>    | $\approx 2.318 \text{ k}\Omega$ (if the values continue as primes)                                                                                                                            | 02 §5             |
| <b>B5</b>    | $I_1 = 5/3 \text{ A}$ , $I_2 = 4/3 \text{ A}$ , <b><math>I = 3 \text{ A}</math></b> (resistors in $\Omega$ )                                                                                  | 03 §1–3           |
| <b>B6</b>    | Mechanical superposition gives 10 mA; the true value is <b>5 mA</b> . Superposition fails for parallel ideal voltage sources                                                                  | 03 §5             |
| <b>B7</b>    | $V_{\text{Th}} = 20/3 \text{ V}$ , $R_{\text{Th}} = 8/3 \text{ k}\Omega$ , <b><math>I_L = 20/11 \approx 1.82 \text{ mA}</math></b> ; Norton: $I_N = 2.5 \text{ mA}$ , same $I_L$              | 04                |
| <b>B8–B9</b> | (a) 10 k $\Omega$ (b) 5 mW (c) 2.5 mW (d) 50 % (e) 5 V (f) $P_L = P_S/2$ , $V_L = V_S/2$                                                                                                      | 05 §2–3           |
| <b>B10</b>   | $R = 5 \text{ k}\Omega$ , $P_{\text{max}} = 5 \text{ mW}$ , $\eta = 50 \%$                                                                                                                    | 05 Check 1        |
| <b>B11</b>   | 0, $V/R$ , $V/R$                                                                                                                                                                              | 06 §1             |
| <b>B12</b>   | 0, 0, $V/R$ for both; $i(t) = 10(1 - e^{-t/1 \mu\text{s}}) \text{ mA}$                                                                                                                        | 06 §2–4           |
| <b>B13</b>   | $I_L(0^-) = I_L(0^+) = 10 \text{ mA}$ . With no path, the ideal circuit has no solution (voltage spike, arc).<br>If the R–L loop closes on itself: $i(t) = 10e^{-t/1 \mu\text{s}} \text{ mA}$ | 06 §5             |